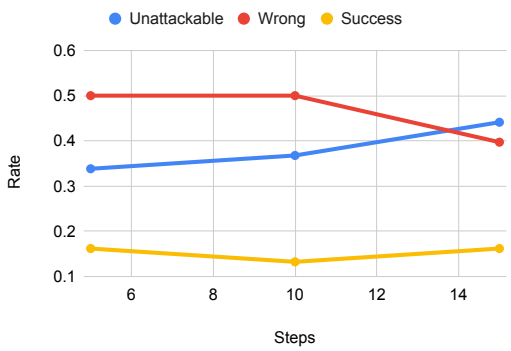
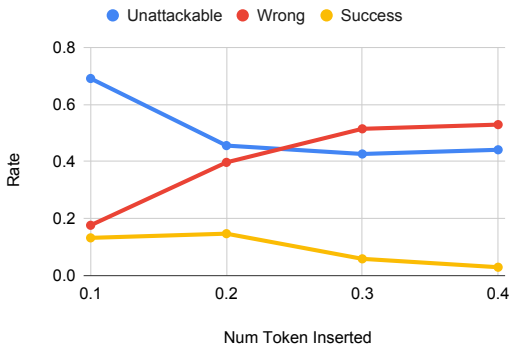
Figure 3: Success examples of our token-level and embedding-level perturbations on different models. We classify the errors into four categories. For token-level perturbations, the replaced tokens are colored in red, and for the CoTs, the wrong steps are colored in red.



(a) Token-level: Number of steps.



(b) Token-level: Inserted token ratio.

Figure 4: Ablation studies on the number of perturbation steps and inserted token ratio.

5.6 Ablation studies

Number of perturb steps. Figure 4a displays the impact of perturbation steps on UR and SR with the GSM8K dataset using the Llama-3-8B model (400 examples): the success rate is generally insensitive to the number of steps, so we choose 5 steps for efficiency purposes for all baselines and MATCHA.

Fraction of inserted tokens compared with the original question. We investigate how different fractions of inserted tokens compared with the original question influence the performance of the Llama-3-8B model on the GSM8K dataset with 400 examples (Figure 4b). From the results, we observe that 0.2 is a good choice to balance both the success rate, efficiency, and the small change in the semantic meaning of the questions.

Additional ablation studies on embedding-level perturbations can be founded in Appendix A.4.

6 Conclusion

In this work, we empirically examine the *Decoupling Hypothesis*, questioning the assumption that correct answers in LLMs necessarily arise from faithful reasoning. Using our proposed MATCHA framework, we introduce *Answer-Conditioned Probing* to disentangle the robustness of a model's prediction from that of its justification. We find that LLMs can often produce correct answers despite fragile or inconsistent reasoning. In particular, small, imperceptible perturbations can disrupt the logical structure of a Chain-of-Thought while leaving the final answer unchanged, especially in multi-step and commonsense tasks. The transferability of these effects to black-box models (e.g., GPT-4) suggests that this phenomenon is not model-specific, but reflects a broader limitation of current training and alignment practices. Overall, MATCHA provides a diagnostic perspective on reasoning robustness, highlighting the gap between answer correctness and process reliability. Our results suggest that improving trustworthiness will require moving beyond answer-level supervision toward methods that better couple reasoning structure with model predictions.

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